



TOXICOLOGY NEWSLETTER

“Research is what I am doing when I don't know what I am doing.”

- Wernher von Braun

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PESTICIDE MAXIMUM RESIDUE LIMITS (MRL'S)

Preamble

For a well-rounded toxicologist, it is imperative that one should have a broad background and understanding in not just the testing part, but also on other factors on food, environment, pesticides, chemicals, dynamics kinetics, biochemical mechanisms, organ effects, risk assessment, use of toxicological data in real life use situations etc. With this objective in mind, I have chosen a topic which is dear and near to me (since I worked for pesticides for nearly two decades), i.e. are pesticides risky to humans and how pesticide manufacture, use, and control is managed by the society at large, through pesticide Maximum Residue Limits (MRL's). Hope, it will be useful to enlarge the background of those who normally do not work in this area.

A. Introduction

Pesticides are chemicals used in agriculture to protect crops against insects, fungi, weeds and other pests. In addition to their use in agriculture, pesticides are also used to protect public health in controlling the vectors of tropical diseases, such as mosquitoes. These pesticides may remain in lesser amounts (called residues) in or on fruits, vegetables, grains, and other foods. But pesticides are also potentially toxic to humans. They may induce adverse health effects including cancer, effects on reproduction, immune or nervous systems. Before they can be authorized for use, pesticides should be tested for all possible health effects and the results should be analyzed by experts to assess any risks to humans. Hence, chemical residues in food are of concern to consumers.

The rapidly growing use of pesticides in agriculture after World War II gave rise to regulation by governments of the sale and use of pesticides to prevent chemicals with unacceptable properties being introduced onto the market. The use of chemicals was regulated to protect the users of pesticides, the consumers of treated foodstuffs, domestic animals and, at a later stage, the environment.

For this purpose, governments requested manufacturers and other data submitters to submit information on the properties of their products and on their intended uses. As differences arose among countries on the extent and scope of data to be supplied, international organizations initiated attempts to harmonize requirements.

To regulate the safety of foods, Regulatory Agencies (including in India – Central Insecticides Board) around the world have set tolerance levels or maximum legal limits for pesticide residues on food commodities for sale. In general, tolerances are based on a very conservative set of assumptions including that each pesticide is applied at the maximum rate allowed by the label, the maximum number of applications are made, and only the minimum permissible interval is allowed between applications. Scientists find the safe daily intake level, "No Observable Effects Level (NOEL)," and build in a 100-fold or more margin of safety. This procedure sets a legal residue level. If the maximum possible exposure to a chemical is less than the legal residue level, the regulatory agencies grant a tolerance.

Tolerances represent the upper limit of pesticide residues and these levels rarely occur in ready-to-eat food commodities. In general, dietary level of most pesticides were less than 1 percent of the Acceptable Daily Intake established by the United Nations Food and Agriculture Organization and the World Health Organization.

It is important to note that farmers generally use pesticides very judiciously. Chemicals are one of the most expensive "inputs" that a farmer can use.

B. United Nations Involvement

Back in the 1950's, when many countries did not have a system in place for establishing MRL's, the United Nations(UN) first initiated a process of establishing MRLs. The UN panel recommended studies to be undertaken jointly by FAO and WHO on the hazards arising from pesticide residues in and on food and feedstuffs, on the establishment of principles governing the setting up of pesticide tolerances, on the feasibility of preparing an International Code for toxicological data and residue data required to achieve the safe use of a pesticide.

This expert group of the FAO Working Party on Pesticide Residues recommended that the following were considered essential:

1. Residue levels resulting from Good Agricultural Practices (GACP) should be obtained by
FAO from governments and pesticide manufacturers. These data should be considered by the FAO Working Party on Pesticide Residues. After consideration of the ADI and of the national nutritional patterns as stated in the FAO Food Balance Sheets, the Working Party would propose tolerances for residues on individual crops for consideration by governments and by the Expert Committee on Pesticide Residues of the Codex Alimentarius Commission
2. Residues found in surveys of marketed commodities
3. ADIs to be estimated by joint meetings of the WHO Committee on Pesticide Residues
and the FAO Committee on Pesticides in Agriculture
4. National nutritional patterns
5. Acceptable analytical methods for residues. These methods should also be adopted by
the Pesticide Committee of the Codex Alimentarius.

C. Why does WHO have 2 Distinct Processes for “Hazard Identification” and “Risk Assessment”?

“**Hazard identification**” - in particular, the IARC classification of substances in terms of their carcinogenicity—is the first step of the “risk assessment” process. Classification of an agent as a carcinogenic hazard is an important indication that some level of exposure, for example from occupation, environment, food, etc., could result in an increased risk of cancer.

Risk assessment for pesticide residues in food, as conducted by the Joint FAO/WHO Meeting on Pesticide Residues (JMPR), establishes a safe intake level after assessing the level of risk. Acceptable daily intakes (ADIs) are used by governments and international risk managers, such as the Codex Alimentarius Commission, to establish maximum residue limits (MRLs) for pesticides in food. MRLs are enforced by national authorities to ensure that the amount of pesticide residues consumers is exposed to through eating food over a lifetime will not have adverse health effects.

IARC’s hazard identification can inform the JMPR’s risk assessment, and thus the two processes can be complementary. For example, IARC may identify new evidence from scientific studies on the carcinogenicity of a chemical and, when necessary, JMPR conducts an evaluation or a re-evaluation of the safety of that chemical as it is used in agriculture and occurs in food.

D. Current Paradigm of MRL Establishment in Regulated Countries

Scientific studies of the potential health effects of hazardous chemicals, such as pesticides, allow them to be classified as carcinogenic (can cause cancer), neurotoxic (can cause damage to the brain), or teratogenic (can cause damage to a fetus). This process of classification, called “hazard identification,” is the first step of “risk assessment”. An example of hazard identification is the classification of substances according to their carcinogenicity to humans.

The same chemical can have different effects at different doses, which depends on how much of the chemical a person is exposed to. It can also depend on the route by which the exposure occurs, e.g. ingestion, inhalation or injection.

That is, the setting of an MRL is driven by two assessments: good agricultural practice (GAP) and the toxicological evaluation of the pesticide in question.

MRLs are thus driven by the toxicological properties of the pesticide in question. They are intended to be sufficiently low so that the consumption of food with pesticide residues at or below the corresponding MRL is safe even for vulnerable consumers.

A pesticide company that seeks approval to use its product in any country is submitting the following two items to the regulatory agency:

1. Toxicological evaluation
2. The residues expected on the produced food items after they have been treated with the pesticide according to good agricultural practice

This toxicology-driven setting of MRLs would imply that substantial amounts of pesticides with a comparatively low toxicity (according to our current toxicological understanding) would be acceptable. To avoid this, MRLs are additionally adjusted to the lowest value that is technically achievable, i.e. to the residues levels that can be achieved when working according to good agricultural practice.

Finally, the MRLs are set to the lowest concentration that can still be determined by state-of-the-art chemical-analytical techniques, if the use of a pesticide according to good agricultural practices would not result in any measurable pesticide residues. The MRL then basically serves as a check on whether farmers and food producers have worked according to best practice.

In case no substance-specific MRLs have been set, a default value of 0.01 mg/kg is used.

E. Measuring Safety: ADI and ARfD

The ADI (Acceptable Daily Intake) refers to the maximum quantity of a substance which can be consumed every day for a lifetime without harm to the consumer. The ARfD (Acute Reference Dose) refers to the maximum quantity of a substance that can be ingested in a single day without any harm to the consumer. This is intended to define (based on all known facts at the time of

the evaluation) an estimate of a chemical substance in food (or drinking water), expressed on a body weight basis, that can be ingested over a short period of time, usually during one meal or one day, without appreciable health risk to the consumer.

The ADI and ARfD are obtained through animal testing and are based on the highest dose where no recognizable harmful effects are observed; the No Observable Adverse Effect Level (NOAEL). In accordance with international practice, the NOAEL is divided by an uncertainty factor of at least 100 to compensate for potential differences between animals and humans – and for differences between individuals. Since the NOAEL may differ for chronic (long-term) and acute (short-term) effects, the ADI and ARfD may be set at various levels. Before an active ingredient can be authorized, a dietary risk assessment is conducted to ensure that the potential chronic and acute exposure of consumers to residues remain below the ADI and ARfD, respectively. Authorization is granted only if the maximum residue levels are shown to be safe under a set of "worst case" assumptions. Consumer protection is ensured through toxicologically based safety limits.

Following flowchart describes the process of establishing the ADI:

F. How to Establish the Acceptable Daily Intake (ADI) per 1 kg Body Weight

This is an estimate of the amount of an agricultural chemical that can be ingested daily over a lifetime without appreciable health risk to the consumer. First, a study is selected in which the adverse effect was observed with the lowest concentration of all the long-term toxicity studies such as chronic toxicity studies in animals. Below is an example of toxicology studies conducted with Insecticide:

NOAELs for an Insecticide in Studies on Rats, Mice, and Dogs

Species	Study	Duration (Weeks)	Route of Administration	NOAEL (mg/kg b. wt.)
Rat	Sub-chronic	13	Feed	2.7
Rat	Reproductive	18	Feed	1.4
Rat	Chronic	104	Feed	1
Mouse	Sub-acute	4	Feed	-a
Mouse	Sub-chronic	13	Feed	8

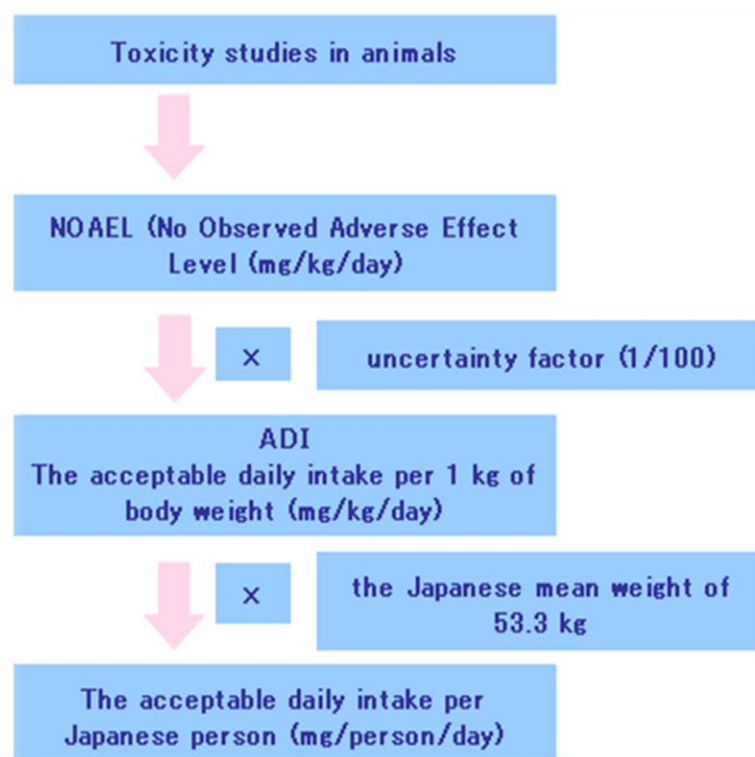
Mouse	Chronic	78	Feed	1.5
Dog	Sub-acute	4	Feed	a
Dog	Sub-chronic	13	Feed	0.9
Dog	Chronic	52	Feed	1

^a The lowest dose was toxic; therefore, no NOAEL was identified

Then the dose level at which no adverse effect was observed in that same study is calculated (NOAEL: no-observed adverse effect level) (mg/kg/day). In the above example of an insecticide, the lowest NOAEL for the chronic study is in a dog which 1 mg/kg/day. Considering that this value is obtained from animal studies and there are variations among individuals, it is multiplied by an uncertainty factor (standard $1/100[1/(10[\text{species difference}] \times 10[\text{personal difference}])]$) to obtain the level that is harmless to humans. The resultant value is the ADI which will be 0.001.

The obtained ADI is an acceptable daily intake per kilogram of body weight, and by multiplying the ADI by the mean weight of 60 kg human which will be 0.06 mg per day per human. This is the value on which the maximum residue level of an agricultural chemical is established.

The following is a flowchart of calculating the acceptable daily intake (ADI)



G. Exposure Assessment

An exposure assessment is necessary in order to reach a conclusion on the acceptability of proposed MRLs and the underlying GAP from a public health point of view. The dietary exposure of a pesticide residue in a given food is obtained by multiplying the residue level in the food by the amount of the food consumed. The total dietary intake of the pesticide residue is then obtained by summing the intakes of all foods containing the residue. Exposure to a pesticide residue present, or likely to be present, in drinking water or in a food for which no MRL has been established should be taken into consideration when such information is available. The estimated dietary intake of pesticide residues resulting from the application of a pesticide and other sources should be less than its established ADI.

H. Determination of the Maximum Residue Level in Crops

The agricultural chemicals sprayed on the surface of the crops decrease over time since the spray day, as they evaporate into the air, wash away due to weather, or decompose due to the interaction of light and water. However, some may remain at harvest.

The maximum residue level in crops is established based on the residual amount obtained from the residue studies in crops, which was submitted at the time of application for registration. Considering that the residue amount fluctuates due to various external factors such as weather conditions, the standards are established, building in a safety factor by making the values more generous than those obtained from the studies, and considering further the foreign standards and international standards (Codex MRLs).

Here is an example of an agricultural chemical that is used for crops such as soybeans, small red beans, sweet potatoes, and others (Table). Given the residue amount of 0.97 ppm for soybeans, 0.87 ppm for small red beans, and 0.47 ppm for sweet potatoes, which were obtained from studies with uniform application methods, each residue value is tentatively established as 2 ppm for soybeans, 2 ppm for small red beans, 1 ppm for sweet potatoes, and so on, all the way down to the last item (strawberries), making the values more generous for safety. Then the estimated intake of agricultural chemicals is calculated based on the resultant values of mean food intake (food factors in the National Nutrition Survey). The total of the estimated intake for each crop is 0.2378 mg, which falls below 80% of the acceptable intake of 4.5 mg. Therefore, the standards in this example are established as 2 ppm for soybeans, 2 ppm for small red beans, and 1 ppm for sweet potatoes.

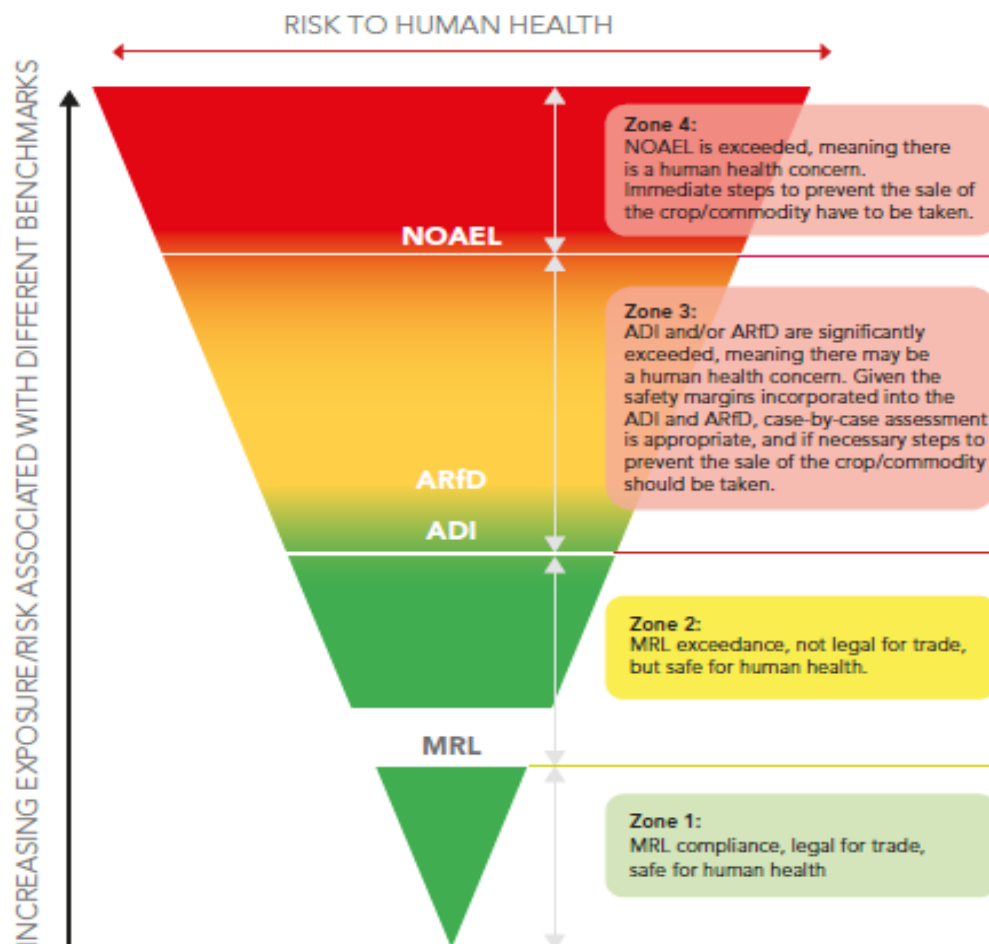
Crops	Application method	Maximum residue amount (ppm)	MRLs	Food factor or (g)	Estimated food intake (mg)	Acceptable daily intake for Adults (ADI× 60)
Soybeans(d ry)	Spraying	0.97	2	56.1	0.1122	
Beans	Spraying	0.87	2	1.4	0.0028	
Sweet potatoes	Spraying	0.47	1	15.7	0.0157	
Beat sugars	Spraying	0.31	1	4.5	0.0045	
Cabbages	Spraying	0.82	2	22.8	0.0456	
Onions	Spraying	0.33	1	30.3	0.0303	
Carrots	Spraying	0.46	1	24.6	0.0246	
Immature kidney beans	Spraying	0.38	1	1.9	0.0019	
Soybeans	Spraying	0.16	0.5	0.1	0.00005	
Strawberries	Spraying	0.15	0.5	0.3	0.00015	
Total					0.2378	<u>4.5</u> <u>mg/person/</u> <u>day</u>

Estimated intake (mg: total of each applied crop [standard value (ppm) × food factor (kg)]) ≤ ADI(mg/kg) ×60 (kg)

I. Risk Assessment

All the above information is summarized in the following diagram

Measuring residue levels



J. Why do MRL Exceedances Occur?

Exceeding the maximum residue Levels does not necessarily imply a risk to health. However, it usually indicates that a pesticide has been incorrectly used. Food products which have residues exceeding MRL cannot be placed on the market. When a farmer uses a pesticide according to the label instructions and Good Agricultural Practices (GAP), the residues in the crop at harvest do normally not exceed the Maximum Residue Level established in the country of use. However, since MRLs are not always harmonized worldwide, MRL exceedances can occur when products are exported to a country with a lower MRL for the specific pesticide and crop combination.

MRL exceedances can occur due to the following reasons:

- A. The crop protection product was not used according to label instructions
- B. The minimum waiting period between the application of the pesticide and harvest was not respected
- C. Incorrect pesticide dosing was used
- D. The crop protection product safety instructions regarding the storage, use, and cleaning of material were not respected
- E. The crop protection product was not registered for the respective country and/or was used illegally;
- F. Authorized pesticides were used in non-authorized commodities
- G. Recent changes in many agricultural practices due to the withdrawal of many pesticides from the market
- H. The food product was imported from a country outside the EU and the use was not covered by a suitable MRL / import tolerance in the EU
- I. Environmental contamination
- J. Change of host country MRLs standards

K. Import Tolerances

An import tolerance is a Maximum Residue Level that is set based on uses registered in foreign countries to allow the import of treated commodities from abroad and facilitate international trade. Import tolerances can be requested, provided specific criteria are met if a trader wishes to import a commodity:

- 1. Containing residues of a substance used in a country but where the commodity is not produced in that country (e.g. EU papayas)
- 2. Treated with a substance no longer or not yet used in the country
- 3. Treated with a substance in use in the country but where the GAP registered in the exporting country is likely to result in higher residues than the host country GAP.

These MRLs are called “import tolerances” and they also must comply with the same high safety standards. Import tolerances facilitate international trade.

L. Is Food Grown Using Pesticides Safe to Eat?

I can say with confidence, in most cases, that the fruits and vegetables we are eating are safer than ever. It is important to note though, that just because a pesticide residue is detected on a fruit or vegetable, that does not mean it is unsafe. Very small amounts of pesticides that may remain in or on fruits, vegetables, grains, and other foods decrease considerably as crops are

harvested, transported, exposed to light, washed, prepared and cooked. The presence of a detectable pesticide residue does not mean the residue is at an unsafe level. In general, the Pesticide Data Program (PDP) detects residues at levels far lower than those that are considered health risks.

M. Are Maximum Residue Levels, Safety Levels?

Maximum Residue Levels (MRLs) are not safety limits: they are set below safety limits. If pesticide residues are found above an MRL, it shows that the user did not use the pesticide according to its label conditions; or follow 'Good Agricultural Practices' (GAP); or the food concerned originated from outside the regulatory limits and is not compliant with the MRLs established.

N. How does United States EPA (and other Agencies) Regulate Pesticides in Food?

EPA evaluates every new pesticide and every new use for safety before registration. Before they may be sold, EPA must ensure that pesticides are safe for human health and the environment when used according to label directions. For each pesticide, EPA evaluates hundreds of different scientific studies.

Through these evaluations, EPA is ensuring the overall safety of proposed pesticide uses as required by FQPA. After pesticide registration, EPA reevaluates its safety every 15 years, taking into consideration any new data.

EPA's process for registering and re-evaluating pesticides is not a closed-door process between EPA and pesticide manufacturers. EPA relies on the best science available and places high value on transparency in decision-making. The public is invited to comment throughout the decision-making process – we request studies and data, take our findings to independent expert panels such as the FIFRA Scientific Advisory Panel, and consult the National Academy of Sciences on broad scientific policy questions. The agency also frequently receives hundreds or even thousands of comments from the public on our draft assessments and proposed decisions.

Public concerns about specific pesticides and food safety do not go unnoticed at EPA. We take incidents of pesticide poisoning and exposure very seriously and look at those incidents as part of our review. EPA can and has used its authority to have products removed from the market immediately when risks are imminent.

While we review dietary exposure to pesticides, we also look at worker exposure and environmental exposure. Risks to workers and the environment can lead to cancelations as well, or restrictions on how and when a pesticide can be used, including, when appropriate, establishing 'no spray' buffer zones to protect the surrounding communities and waterways.

Before allowing the use of a pesticide on food crops, EPA sets a maximum

legal residue limit (called a tolerance) for each treated food. The tolerance is the residue level that triggers enforcement action. That is, if residues are found above that level, the commodity will be subject to seizure by the government. EPA receives information on how much pesticide residue remains on various foods through the PDP. Through annual sampling, PDP has collected thousands of samples on 10-15 food commodities and can detect residues at levels far lower than those that pose health risks.

In setting the tolerance, EPA must make a safety finding that the pesticide can be used with "reasonable certainty of no harm." To make this finding, EPA considers the toxicity of the pesticide and its breakdown products, how much of the pesticide is applied and how often, and how much of the pesticide (i.e., the residue) remains in or on food by the time it is marketed. EPA ensures that the tolerance selected will be safe. The tolerance applies to food grown in the U.S. and imported food.

O. An illustrative example: Bifenazate MRL in Peaches

The agency has established an ADI of 0.01 mg/kg b.wt. per day for bifenazate and an ARfD of 0.1 mg/kg b.wt.. The MRL for peaches is 2 mg/kg. Toxicologically, the NOAELs (and LOAELs) in mice, rats, and dogs are similar over a broad range of exposure durations as table below. This indicates that bifenazate has a similar potency over a broad range of exposure durations, as the toxic dose levels derived after a few exposures and after chronic exposures are similar.

NOAELs for Bifenazate identified in studies on Rats, Mice and Dogs

Species	Study	Duration (Weeks)	Route of Administration	NOAEL (mg/kg b. wt.)
Rat	Sub-chronic	13	Feed	2.7
Rat	Reproductive	18	Feed	1.4
Rat	Chronic	104	Feed	1
Mouse	Sub-acute	4	Feed	a
Mouse	Sub-chronic	13	Feed	8
Mouse	Chronic	78	Feed	1.5
Dog	Sub-acute	4	Feed	a
Dog	Sub-chronic	13	Feed	0.9
Dog	Chronic	52	Feed	1

^aThe lowest dose was toxic; therefore, no NOAEL was identified

According to the WHO food consumption, the chronic peach consumption for a 16.15 kg child is estimated to be 2.2 g/day. One peach on average weighs 127.6 g, and therefore a day's portion of 2.2 g is only 1.7% of a single peach. The resulting chronic exposure estimate is 2.7% of the ADI. The regulatory conclusion here is that the MRL of 2 mg/kg for Bifenazate in peaches is safe because the chronic daily intake of peaches—i.e. 1.7% of a single peach assigned an MRL of 2 mg/kg Bifenazate—is of no health concern.

P. Conclusion

The regulatory system for pesticides aims to ensure that the proper use of authorised products carries no unacceptable risk to people or to the environment. Central to our regulatory controls on pesticides is the prior authorisation procedure whereby authorisation must be given before a pesticide can be sold or used. Issues such as pesticides residues are examined as part of the pesticides authorisations process. Just like medicines, crop protection products are subject to regulations. The use of pesticides is authorized only after an independent expert risk assessment has checked that any residues remaining after the correct use of the product will not lead to any consumer concern. Farmers must comply with Good Agricultural Practices(GAP), following the basic principle of using pesticides as little as possible and only when necessary. We expect that the use of authorised pesticides will lead to lesser amounts of pesticides (residues) on food. Residues are also likely to end up in the environment and in drinking water. When an agency authorises a pesticide, it gives conditions of use which are written on the label. These conditions make sure that any pesticide residues in food will be below levels which could cause harm. We also set legal limits on residues in food: these are called Maximum Residue Levels (MRLs).

The potential residues on a harvested crop are regulated by a maximum level (MRL) which is set As Low As Reasonably Achievable; the ALARA principle. The MRL is an important trading standard. Maximum Residue Limits (MRLs) for pesticide residues are the maximum concentrations of residues to be legally permitted in or on a food. MRLs for pesticides may also be applicable to animal feeds. MRLs are set by the Codex Alimentarius Committee (CAC), acting as the risk manager. Draft MRLs for adoption by the CAC are elaborated by the relevant Codex committees, the Codex Committee on Pesticide Residues (CCPR) based on scientific expert advice provided by the risk assessor—i.e. Joint Meeting of Pesticide Residues (JMPR). The scientific advice developed by JMPR aims to provide maximum residue levels for individual crops, plant, and animal products, based on the results of scientific studies so that these levels can be used by the relevant Codex committee to develop the draft MRLs, which may be adopted by the

CAC.JMPR evaluates residue data to estimate likely maximum residue levels in food commodities resulting from pesticide use according to GAP, i.e. with pesticide use for effective pest control, but leaving a residue that is the smallest amount practicable. The use must be safe for the user and the environment, and residues must be safe for the consumer.

Questions

1. *How to establish the Acceptable Daily Intake (ADI) per 1 kg Body Weight?*
2. *Why does WHO have 2 Distinct Processes for “Hazard Identification” and “Risk Assessment”?*
3. *What is ADI and ARfD?*
4. *Why do MRL Exceedances Occur?*
5. *What is meant by Import Tolerances?*
6. *How does United States EPA (and other Agencies) Regulate Pesticides in Food?*

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